

VCF-RDFizer: From a VCF File to Explicit, Verifiable, and Governable Semantic Genomic Data

Elias Crum^{ORCID}^{1,2*}, Bart Buelens^{ORCID}¹,
Ruben Taelman^{ORCID}², Gökhan Ertaylan^{ORCID}¹

²IDLab, Department of Electronics and Information Systems, Ghent University – imec, Technologiepark-Zwijnaarde 116, 9052, Belgium.

¹Digital Biological Systems, Flemish Institute for Technological Research (VITO), Boeretang 200, 2400, Belgium.

*Corresponding author(s). E-mail(s): elias.crum@ugent.be;

Abstract

Background: The Variant Call Format (VCF) is widely used, compact, and readily parsable, but integrating its observations with changing biomedical evidence, clinical context, and data-use policies requires relationships beyond the file itself. Making these relationships explicit could support variant reanalysis, pharmacogenomic decision support, and governed data reuse, provided that the converted information remains checkable against its source. Here we present VCF-RDFizer, a command-line tool that converts VCF into the Resource Description Framework (RDF) using the public VCF Core vocabulary, as a first step toward these applications.

Results: We evaluate VCF-RDFizer using public genomic datasets, source-based validation, scaling experiments, and a linked workflow combining genomic observations with external knowledge and simulated access policies. The results show that VCF data can be converted into an explicit RDF representation while retaining the information needed to check the transformed data against its source. Complementary query- and constraint-based validation expose different classes of transformation errors, demonstrating the value of validating both data content and graph structure. The resulting RDF representations also support queries that combine genomic observations with external variant knowledge, population information, and requester-specific policies while reproducing the results of an equivalent conventional VCF workflow. These capabilities come with additional

conversion and indexing costs: selected repeated queries benefited from an existing graph index, whereas direct VCF parsing remained advantageous for the tested one-pass analyses and for small region queries on a tabix-indexed VCF.

Conclusions: VCF-RDFizer supports verifiable conversion and reusable, policy-aware research queries; it is not a universal replacement for VCF. Its value depends on the relationships needed, the validation coverage achievable, and the cost of conversion and indexing.

Keywords: Genomic variation, Variant Call Format, RDF, SPARQL, Semantic validation, Data-use policies

1 Background

The Variant Call Format (VCF) is widely used to represent and exchange genomic variation in research and clinical sequencing, including in pharmacogenomics and precision medicine [1–3]. It is compact, readily parsed, and already supports identifiers, reference information, and extensible annotations [1, 4]. However, many questions in healthcare and biomedical research require more than the variants recorded in a file. Practical uses combine genomic observations with clinical records, external biomedical knowledge, sample metadata, and conditions governing data use [5–7]. The challenge is therefore not simply to read VCF, but to make the relationships needed for downstream uses explicit and reusable across independently maintained resources. Three generalized use cases illustrate such needs.

*****(1) Clinical variant interpretation and reanalysis.*****

Clinical interpretation combines genomic observations with clinical assertions, population frequencies, gene–disease associations, and patient phenotypes, all of which can change over time [8, 9]. Reanalysis can therefore identify new pathogenic or likely pathogenic findings and downgrade earlier classifications [10]. The observation and the evidence used to interpret it have different lifecycles. Keeping the relationship explicit and reusable between observation and the evidence is difficult when this interpretation is stored only in annotations or application-specific databases.

*****(2) Pharmacogenomic clinical decision support.*****

Pharmacogenomic decision support connects genomic observations to an interpreted phenotype, a medication, an indication, and clinical guidance. For example, CYP2C19-guided clopidogrel recommendations depend on both metabolizer phenotype and clinical indication [11]. Current systems still face challenges in representing genomic results as discrete data, integrating them with electronic health records (EHRs), and maintaining the knowledge used for interpretation [12, 13]. These relationships span systems and must remain traceable if genomic observations are to support changing clinical decisions.

***** (3) Consent-aware genomic data reuse. *****

Secondary use adds a governance requirement to these integration problems. Data access decisions must reflect the uses permitted by participant consent, which can differ across participants, datasets, and purposes [7, 14]. The GA4GH Data Use Ontology (DUO) and Machine Readable Consent Guidance provide machine-readable terms for expressing these conditions [7, 14, 15]. Because genomic data are highly sensitive [16, 17], selective release of participants, regions, or variants must remain tied to the conditions that govern their use.

None of these tasks is impossible with the current VCF representation. Variant-analysis frameworks, clinical systems, and data-access infrastructures already address important parts of these workflows [7, 12, 18]. The remaining challenges increasingly concern how these systems interoperate, how computational and operational costs are balanced, and how analyses can be reproduced as external knowledge and policies change. VCF itself is not designed to provide a shared representation linking a source observation to evolving evidence, patient context, and the conditions governing a particular request. These relationships are therefore commonly reconstructed in annotations, local databases, or application logic surrounding the file(s). As a result, there have become many disparate, narrowly scoped solutions for connecting variant data to external data and, thus, motivates a shared representation in which these relationships can be expressed explicitly, reused across applications, and queried independently of the source file(s).

The Resource Description Framework (RDF) provides one way to realize such a representation. RDF expresses relationships as triples whose resources can be identified by Internationalized Resource Identifiers (IRIs) [19, 20]. Linked Data practices use shared identifiers to connect independently managed resources, while SPARQL provides a language for querying those relationships [21–25]. For genomic variants, NCBI’s Sequence Position Deletion Insertion (SPDI) data model provides such a shared identity, so that the same allele reported in different files or in resources such as dbSNP and ClinVar can be recognized as one [26]. This allows an observation in a source file to remain distinct from the variant it describes, the evidence used to interpret it, and the policies governing its use. New evidence, prescribing guidance, or data-use conditions can therefore be represented as separate statements and queried alongside the observation without rewriting the source file or re-encoding the relationship for each application. The Open Digital Rights Language (ODRL) can similarly express policies over identified resources [27]. Realizing these benefits still depends on well-defined variant identities, consistent interpretation rules, and policy-evaluation mechanisms that act on the represented data [27, 28]. RDF therefore provides the representational foundation for these workflows, while normalization, clinical interpretation, and access enforcement remain responsibilities of the systems built around it.

The potential of RDF for VCF representation has already motivated several semantic-genomics initiatives and VCF-to-RDF converters (Table 1). Many of these tools were developed for focused applications or as components of larger pipelines, and consequently represent only the VCF information required for those tasks [29–33]. Existing tools also differ in their RDF models, treatment of header and genotype information, and mechanisms for extending the resulting representation [30, 32, 34–36].

The limited preservation of source context and single predefined application scoping of existing VCF-to-RDF approaches leaves an open need for a reproducible conversion that supports more generalized linking, validation, and downstream reuse. Non-RDF frameworks such as Hail, GEMINI, and TileDB-VCF, and standards such as Beacon, htsget, and FHIR Genomics, address complementary parts of variant analysis, retrieval, and clinical exchange but do not directly address the challenge of connecting VCF data to external data reliably [18, 37–41]. The goal of an RDF representation is therefore not to replace these systems, but to provide a reusable semantic layer through which genomic observations can be connected to them and to other external resources.

Table 1 Documented capabilities of VCF-to-RDF converters and related workflows, from public documentation inspected in September and October 2026, not measurements on a shared input.

Tool or workflow	Representation			Operation			
	RDF model	Header + genotype	Editable mappings	CLI	Reproducible workflow	Compressed RDF	Public release
<i>Dedicated VCF-to-RDF converters</i>							
VCF-RDFizer (this study)	●	●	●	●	●	●	●
TogoVar VCF2RDF [32]	●	○	●	●	●	○	●
JVarkit VcfToRdf [30]	●	●	○	●	●	○	●
SPARQLing Genomics [34, 35]	●	●	○	●	●	●	●
VCF2RDF (web) [42]	●	●	○	●	●	○	●
BioHackathon vcf2rdf [29, 43]	●	●	○	●	●	○	●
<i>General genomic-format tools</i>							
BioInterchange [36, 44]	●	●	●	●	●	○	●
bio-vcf [31]	○	●	●	●	●	○	●
ruby-rdf/rdf-vcf [45]	○	○	○	●	●	○	●
<i>Knowledge-graph construction pipelines</i>							
VariantKG [46]	●	●	○	●	●	○	●
AgroLD VCF2RDF [47]	●	○	○	○	○	○	●
<i>Virtual semantic access</i>							
Semantic Beacons [33]	●	○	●	○	●	○	●

● supported and documented ● partial or limited ○ not found in public documentation. Supplementary Table S1 defines each column and gives the evidence for every mark.

For these applications, producing RDF is only a first step. Its usefulness also depends on whether the representation can be checked against the source and whether the costs of producing and using it are practical. Biomedical semantic tools such as SPHN Schema Forge, FAIR-Checker, and a FHIR RDF enrichment pipeline provide useful precedents for evaluating transformations alongside the validation and information requirements of their intended workflows [48–50].

We present VCF-RDFizer, a tool that converts VCF into RDF using the public VCF Core vocabulary and evaluates the resulting representation [51]. The vocabulary was developed separately; here, we focus on the representation choices needed to assess the converter and its downstream use [52]. We evaluate three capabilities together: source-verifiable conversion, linking to external knowledge maintained separately from the genomic data, and policy-aware release. A linked research workflow combines genomic observations, ClinVar classifications [53], population frequencies, and simulated data-use policies, with an equivalent conventional VCF workflow as its comparator. The experiment captures shared requirements of the three motivating use cases without claiming to implement clinical reanalysis, pharmacogenomic decision support, or a secure data-access service. Three questions organize the evaluation: **RQ1**: Which aspects of VCF information are preserved and verifiable? **RQ2**: What do the explicit VCF field representations enable in a linked, policy-aware workflow? **RQ3**: Under which workloads and representation choices are the costs of conversion, evaluation, linking, and querying practical? Together, these experiments evaluate whether VCF can serve as the basis for a reusable semantic representation that remains verifiable against its source, can be connected to external information, and provides a practical foundation for the broader use of VCF data.

2 Implementation

VCF-RDFizer¹ is a command-line tool that converts VCF files into RDF and validates the result against its source (Figure 1). The tool is available through PyPI [54], conda-forge [55], and Docker Hub [56], and runs its conversion toolchain in a single pinned Docker image, so that results do not depend on locally installed versions. The tool offers various functionalities including conversion, compression, decompression, validation, data linking, and policy attachment with documentation, usage guides, and examples explained extensively in the tool’s [README.md](#).

2.1 Representation of VCF

VCF-RDFizer generates a knowledge graph that represents the source VCF document, not only an interpreted variant. The conversion is built using the VCF Core Vocabulary [51, 52]² whose RDF output separates file and header assertions, variant records, and sample/genotype assertions (Figure 2). Supplementary Figure S1 gives the complete model, and the vocabulary itself is described separately [52]. The vocabulary aligns with FALDO for coordinates, the Sequence Ontology for variant types, HERO for genomic constructs, and features of the GA4GH Variation Representation Specification (VRS) [57–61]. These alignments connect VCF-specific structure to existing models; they do not replace the need to preserve the source’s interpretation context.

Deterministic IRI templates identify resources by source file and record position. The same biological alteration in two files therefore remains two observations, each with its own quality values, annotations, and provenance. A separate linkset can connect both to a shared variant identifier (Supplementary Figure S2). File-scoped

¹Source code and implementation repository: <https://github.com/ecrum19/VCF-RDFizer>.

²Vocabulary documentation: <https://ecrum19.github.io/vcf-core-vocabulary/>.

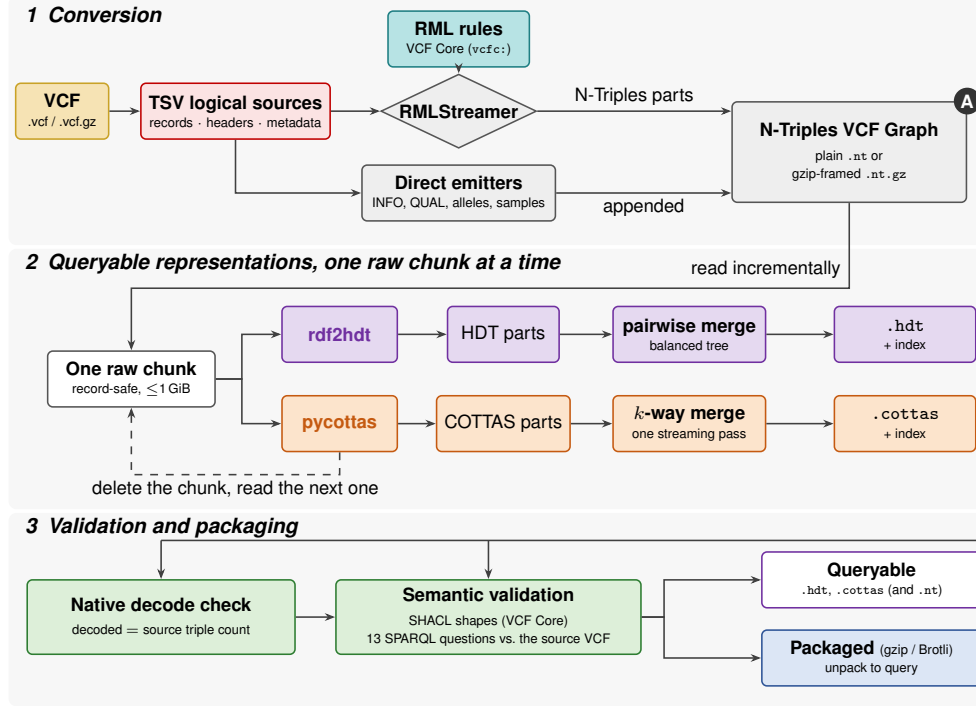


Fig. 1 VCF-RDFizer v3.1.0 implementation. RML mappings and direct emitters produce the source graph, which is appended incrementally in plain or gzip-framed N-Triples. Optional HDT and COTTAS construction uses one raw chunk at a time, followed by merging and indexing. Decoding/count checks against the N-Triples VCF graph (A), source comparisons, and enabled shape profiles assess different properties. Gzip and Brotli packages serve transport or archival use; their compression is distinct from a queryable format.

observation identity and cross-file variant identity serve different purposes and are not collapsed.

Two sample profiles offer different access granularity. *Expanded* makes sample calls and their FORMAT values directly addressable resources, whereas *condensed* retains them in ordered vectors that require decoding. Both preserve file, header, and record context. The different sample representation profiles primarily impact the conversion of single-sample VCF files versus cohort-scale multi-sample VCF files. The choice affects how individual sample values can be queried, not only the size and number of triples in the resulting RDF representation. Supplementary Section S2 describes the representation model and scaling derivations in greater detail.

2.2 Conversion and validation

A conversion run first rewrites each VCF as tab-separated tables, then maps them to RDF and, optionally, builds compressed representations and validates the RDF result. RDF Mapping Language (RML) mappings executed by RMLStreamer produce the record backbone, while direct emitters supply the remaining typed values,

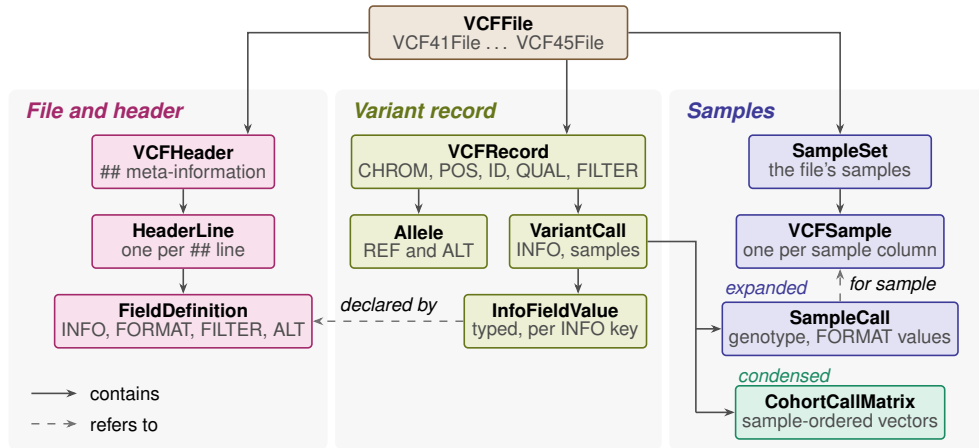


Fig. 2 Central terms of the VCF Core Vocabulary model emitted by VCF-RDfizer. A versioned file contains its header, records, and sample set; a record contains its alleles and its call, which holds INFO values and, per sample, either sample calls (expanded profile) or sample-ordered vectors (condensed profile). Values refer back to the header declarations that define them. All terms use the `vcfc:` prefix; Supplementary Figure S1 adds subclasses, value items, genotypes, and property names.

header attributes, and sample structures [62, 63]. The implementation is therefore hybrid: changing the mappings alone does not change the complete representation. Output is appended incrementally to plain or gzip-framed N-Triples. HDT, COT-TAS, and additional gzip or Brotli packaging of RDF outputs are optional artifacts, not prerequisites for obtaining or querying RDF [64–66]. Native decoding validation compares the triple count of each HDT or COTTAS output with that of the source graph, the N-Triples VCF graph produced by conversion (A in Figure 1). Supplementary Section S2.2 describes the emitters, chunk construction, and query paths in more detail, including the distinction between native compressed access and QLever’s separate index [67–69].

Validation, run after conversion or on existing RDF, compares thirteen SPARQL results with corresponding answers derived independently from the source VCF using `cyvcf2` and, for exact FILTER text, `bcftools` [4, 70, 71]. The comparisons cover record and genotype summaries, metadata and graph inventories, and summaries of values associated with their source records. After normalization, corresponding results are compared for exact agreement, although the value summaries do not constitute exhaustive value-by-value comparisons. Preflight queries and Shapes Constraint Language (SHACL) constraints assess additional structural properties of the graph [72]. The default shapes check cardinality and datatypes, while deeper profiles assess uniqueness and value agreement. Table 2 summarizes the validation coverage for each part of the representation and its limits. The complete query definitions, normalization rules, shape profiles, and execution limits are provided in Supplementary Section S3.

Table 2 Representation contract and assessment boundary. Supplementary Table S2 gives the complete capability matrix; Q1–Q13 are the source-comparison queries (Supplementary Table S3) and shapes the Shapes Constraint Language (SHACL) constraints (Section 2.2).

Information	Representation	Checked by	Limit
File and header	Versioned file, declarations, and linked metadata	Q7–Q10, shapes	Attribute–value agreement needs deeper shapes
Record and alleles	Core fields, ordered alleles, typed and explicitly missing values	Q1–Q4, Q11, shapes	A summary digest, not exhaustive equality
INFO	Raw string plus optional typed, structured values	Q12, structural validation	Preserved, not interpreted
Samples and FORMAT	Expanded sample calls or condensed vectors	Q5, Q6, Q13	Condensed values need decoding
VCF versions	4.1–4.5 file classes; newer and symbolic fields retained	Fixtures	Partly interpreted; local alleles unresolved
Links and policies	Separate linksets; policy-defined release views	Use case	Identity assumptions and simulated policies

2.3 Linking and policy extensions

Two plug-ins extend the VCF RDF graph without modifying the converted source representation. `vcf-rdfizer-link` executes linkers declared in Turtle, with each linker defining a join strategy, reference resource, and emitted relationship. The plug-in writes the resulting links to a separate linkset and reports coverage and skipped records. In the evaluated workflow, linkers connect eligible alleles to SPDI identifiers and variant calls to overlapping genes. The SPDI linker harmonizes contig names and minimally trims explicit alleles, supporting consistent joins under the tested normalization assumptions without claiming canonical identity across all equivalent variant representations [26].

`vcf-rdfizer-policy` evaluates ODRL rules over files or declared graph selections, including genomic regions, named variants, and gene-linked records [14, 27]. The implemented policy profile applies default-deny and deny-overrides semantics, with permitted purposes drawn from a subset of DUO. Ownership rules define which dependent resources are removed when a file or record is withheld. For each requester, the plug-in materializes a release view, records which data were released or withheld, and validates the view against both the policy and an oracle derived from the source VCF text. This evaluation demonstrates policy-aware release under simulated rules, while anonymization, obligation enforcement, and deployment-level security remain outside the scope of the study. Supplementary Section S4 describes the linking tiers, policy evaluation, and release-validation procedure in detail.

3 Evaluation

Three experiments address the research questions: representation fidelity and validation (RQ1), a linked, policy-governed genomic workflow (RQ2), and scalability and retrieval (RQ3). Unless stated otherwise, the RQ1 and RQ3 measurements come from one set of benchmark runs of the published VCF-RDFizer v3.1.0 release, which we call the *base campaign*: 143 runs in 12 experiment families covering conversion and scaling, storage and artifacts, difficult inputs, configuration coverage, validation, and query cost (Supplementary Section S5.1). The linked workflow, the regional and large-graph retrieval runs, and two additional validation runs ran separately and are reported separately.

3.1 Representation fidelity and validation

The public corpus spans five Personal Genome Project donations, an HG002 precisionFDA/Genome in a Bottle (GIAB) challenge file, the 2,504-sample 1000 Genomes chromosome-20 call set, HGSVC2 structural variants, and the HG004 and HG005 GIAB v4.2.1 benchmark VCFs [73–77]. The corpus experiment processes the first 250,000 records of eight inputs and the complete HG005 VCF, meaning all 3,856,856 of its records on chromosomes 1–22; the large cohort is excluded from the expanded-profile corpus run and assessed through controlled slices. The inputs are listed in Supplementary Table S12.

Paired validation covers the difficult-input fixtures, a 10,000-line (9,986-record) single-sample fixture under both sample profiles and three RDF artifacts, and slices up to 100,000 HG005 records. After the base campaign, a separate *consumer WGS validation run* applied the paired validation and the default shapes to the first 250,000 records of NG131FQA1I, a consumer whole-genome sequencing (WGS) VCF, producing a larger graph than any validated in the base campaign. A fault-injection experiment tests 113 deliberate corruptions with queries alone, with the default shapes, and with all shape profiles; unmutated controls test whether the validator rejects valid inputs. Separate large-graph retrieval runs compare the same thirteen answers, without SHACL, on graphs converted from one million HG005 records and from the complete HG005 VCF. The specification, exclusions, and oracle dependencies are in Supplementary Section S3.

3.2 Linked genomic workflow

*****Scope.*****

The three use cases introduced in Section 1 share a common requirement: genomic observations should remain traceable to their source while being linked to information maintained elsewhere. We therefore evaluate a single linked workflow that combines source observations with external evidence and requester-specific release rules, rather than implementing any of the three applications in full. Agreement with a conventional workflow tests whether this linked, policy-aware route reproduces the defined results. The separate frequency graph and applied rule changes then test whether external information and release conditions can be reused without modifying the

converted VCF RDF representations. Together, these experiments assess core prerequisites of the motivating applications, while clinical effectiveness, production security, and application-specific deployment remain outside the scope of this study. Supplementary Section S4 maps each motivating use case to the capabilities tested here and the additional work required for implementation.

*****Application question.*****

The application asks: *which observed variants in the fixed 81-gene ACMG SF v3.2 secondary-findings set have a corresponding ClinVar classification, which participant and gene are associated with each match, and which matches may be released to each requester?* The benchmark fixes the gene set to ACMG SF v3.2 for reproducibility and does not represent the latest ACMG recommendations [78]. ClinVar variants are included when they have at least one review star, regardless of classification. Pathogenic and likely pathogenic (P/LP) matches are reported separately. These P/LP matches are not treated as reportable secondary findings because the workflow does not evaluate the additional gene-, inheritance-, and variant-specific criteria required for clinical reporting.

*****RDF and conventional comparison workflows.*****

The RDF workflow links the converted VCFs to ClinVar and gene resources, materializes a release view for each requester, and queries each view with SPARQL. The conventional workflow harmonizes contig names, annotates variants with bcftools, and evaluates the same case definition in a separate Python implementation. The two workflows are compared using tuples defined by participant, gene, contig, position, and alleles. Timings are reported only when the sorted tuple sets agree exactly. Reported counts therefore represent participant-variant-gene matches rather than unique participants. Both workflows share preprocessing and the case definition, while the downstream linking, policy evaluation, and query implementations remain separate.

*****Evaluation arms and simulated release policies.*****

We define an *experimental arm* as a distinct dataset and policy scenario used to test one aspect of the linked workflow. Four arms evaluate integration, cohort breadth, scale beyond the gene spans, and fine-grained release. Arm 1 contains the ACMG gene-span regions of five single-sample VCFs from three sources and tests integration across heterogeneous inputs. Arm 2 contains 104 unrelated high-coverage 1000 Genomes participants, four from each population, and tests the workflow across a larger cohort [79]. Arm 3 processes the complete HG005 VCF, without the gene-span restriction, to test scale. Arm 4 processes the complete NB72462M consumer WGS VCF with policies that produce different non-empty release views from the same file.

Arms 1 and 2 use normalized, split records restricted to the ACMG gene spans. For Arm 2, repeated panel-level INFO is removed from individual participant files and the required population frequencies are represented once in a separate sites-only graph. The Arm 3 result is also compared with the HG005 result from the restricted Arm 1 input. All arms use the ClinVar release from 2026-09-13 [80].

All consent and access conditions are simulated. The first three arms model requesters for clinical care, disease-specific cardiovascular research, and general research use. File-level permissions, participant withdrawals, and a restriction covering 28 cancer-predisposition genes determine the resulting release views. Arm 4 adds the participant’s physician and combines purpose-specific gene-panel, genomic-region, and single-variant rules so that each requester receives a different subset of the same VCF’s records. Each participant is represented by a single-sample VCF, so the experiment does not assess selective masking of samples within a multi-sample VCF. Four additional rule changes test where policy configuration or execution logic changes in each workflow; they do not measure usability or maintainability. Supplementary Section S4 provides the complete arm definitions, policy rules, and evaluation protocol.

3.3 **Scalability and retrieval**

We use controlled scaling *ladders* to measure how conversion costs change as dataset size increases along one dimension at a time. The *record ladder* processes progressively larger prefixes of HG005 containing 10,000, 100,000, and one million records, each with three replicates, followed by a single run on the complete HG005 VCF. The *sample ladder* holds 10,000 cohort records and their INFO content fixed while increasing the number of samples from one to 2,504 under both sample-representation profiles. For each ladder, we measure stage timings, resident memory, transient workspace, and retained artifact sizes separately. Measurements on the complete HG005 VCF and on the whole corpus are reported as single runs rather than replicated estimates.

Retrieval experiments compare individual questions, the complete thirteen-question workload, and five regional queries in QLever against parsing the VCF with *cyvcf2* or *bcftools*, either in full or through region queries on a *tabix* index. Regional queries use windows from 1 kb to 10 Mb, with twenty windows per size and three replicates. All execution routes apply the same POS-based record-selection rule so that timings are compared only after equivalent results are established. Indexing and other setup costs are reported separately from query execution.

All benchmarks ran on three dedicated hosts with the same 8-core Intel Xeon Silver 4309Y processor and 33.6 GB of RAM without swap: two identical hosts ran the base campaign and Arms 1–3 of the linked workflow, and a third ran the large-graph retrieval runs and Arm 4. Each experiment ran alone on its host inside a pinned Docker image, and timings are compared only within a host. Supplementary Section S5 lists the experiment families of the base campaign and the separate runs, with their software versions, hosts, exclusions, and archival status. All experiments and analyses can be reproduced from the public *vcf-rdfizer-testing* repository³, which contains a script for each experiment, the scripts that download and derive the inputs, the analysis and figure code, and the archived results behind the reported numbers.

³<https://github.com/ecrum19/vcf-rdfizer-testing>

4 Results

4.1 RQ1: Source comparisons and structural validation provide complementary evidence

Figure 3 summarizes the observed validation outcomes against the expected result for a faithful conversion and a validator capable of detecting disagreement. Of the 78 validation runs in the base campaign, 76 completed, drawn from five experiment families: the difficult-input fixtures, the configuration covering set, the cross-encoding fixture, the operating-mode checks, and the query-cost slices of up to 100,000 HG005 records (17.1M triples). Across these runs, 984 SPARQL results had corresponding oracle results derived from the source VCF, and all 984 pairs agreed exactly after normalization. Four additional query instances were not applicable: the zero-record fixture contains no genotypes, so genotype queries Q5 and Q6 have no defined result for either of its two RDF artifacts. All 1,144 internal consistency checks also passed, including relationships between independently calculated summaries, such as the requirement that transition and transversion counts sum to the number of biallelic single-nucleotide variants.

To test whether these results depended on a particular SPARQL implementation, 25 of the 76 validation runs were repeated on four SPARQL engines. All four engines returned identical normalized answers for every query in those runs; the remaining 51 runs were executed with a single engine. Serialization fidelity was assessed separately for 152 HDT and COTTAS artifacts. After decoding, every artifact contained exactly the same number of triples as its source RDF graph.

Each validation layer tests a different property of the conversion. Query agreement establishes equality for the specified summaries, while SHACL validation establishes only the constraints enabled by the selected profile; neither demonstrates complete semantic equivalence between VCF and RDF. Decoding checks confirm triple counts after serialization, not value-by-value identity. The retrieval run on the complete HG005 VCF compared query answers without SHACL, and neither it nor the consumer WGS validation run contributes to the 984 comparisons reported for the base campaign.

In the consumer WGS validation run, the first 250,000 records of NG131FQA1I produced 58.2 million triples. All thirteen query results equaled their source-derived values, and the default shape profile, validated in record batches, reported no violations (Supplementary Section S3.2).

Of the 113 injected faults, the queries alone detected 96 (Figure 3b). The other 17, in ten fault classes, changed relationships or values that no query summary inspects, such as which allele a genotype refers to, a sample’s index, phasing, and header attribute values. The default shape profile detected none of these 17; adding the two deeper profiles detected all of them, and both unmutated control graphs still passed. On the fault-injection fixture’s graph of about 2,000 triples, the two deeper profiles took 33.7 and 73.2s, against 1.7s for the default profile. The default profile also ran in 58 base validations, on graphs of up to 0.96M triples from the 10,000-line fixture, and found no violations.

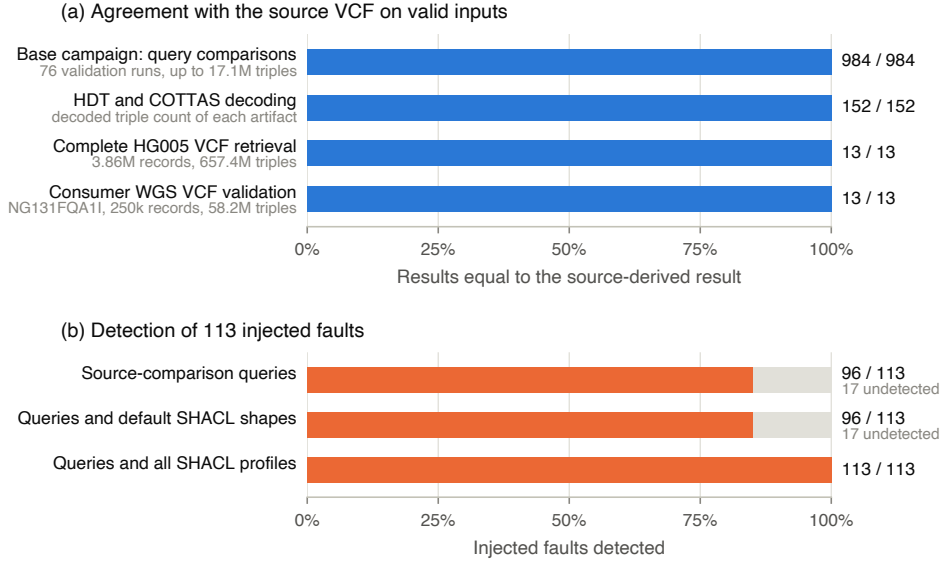


Fig. 3 Validation outcomes on valid inputs and on injected faults. Each bar spans the expected outcome and is filled to the observed one. (a) Results equal to those derived independently from the source VCF: the 984 applicable query comparisons of the base campaign’s 76 validation runs, the decoded triple count of each HDT and COTTAS artifact, and the thirteen queries of the separate retrieval run on the complete HG005 VCF and of the consumer WGS validation run. (b) Injected faults detected by the source-comparison queries alone, with the default SHACL shapes, and with all shape profiles.

During development, this validation found three real defects, all since fixed. The predicate and class censuses (Q9 and Q10) showed that a header-only file omitted the samples its header declares; the shapes reported genotypes that referred to allele resources a raw-INFO conversion never emitted; and the policy plug-in’s own check found a withheld file’s header surviving in release views. A record count alone would have shown none of them. Version fixtures (VCFv4.1–4.5, and an undeclared version) converted, and nine of eleven difficult inputs converted and validated; a truncated gzip was refused, while a malformed sites-only fixture left that paired-coverage test incomplete. Supplementary Sections S6, S3, and S7 retain these outcomes, repeatability validation, and unresolved oracle limitations.

4.2 RQ2: Shared identifiers enabled reusable joins and verified, requester-specific views

Across all four arms, the RDF and bcftools workflows returned identical requester-specific match sets (Figure 4). Every materialized release view also passed the policy and source-validation checks. For HG005, the query over the complete VCF in Arm 3 reproduced the result from its gene-span slice in Arm 1 exactly, returning 1,496 participant–variant–gene matches. These counts represent match tuples rather than unique participants. The unrestricted pathogenic/likely pathogenic (P/LP) subset

contained no matches in Arms 1, 3, and 4 and eight in the Arm 2 cohort. These are ClinVar classifications retrieved under the study definition, not clinically adjudicated secondary findings.

Arm 4 extended the evaluation to fine-grained release from the complete NB72462M VCF, 5,063,417 records after normalization. The participant’s physician received every record, clinical care received all but six APOE records, general research received all but 2,520 records covered by the cancer-gene, APOE, and single-variant restrictions, and disease-specific research received only the 6,397 records within its cardiovascular panel. The RDF and conventional workflows agreed both on the released-record counts and on the resulting variant matches.

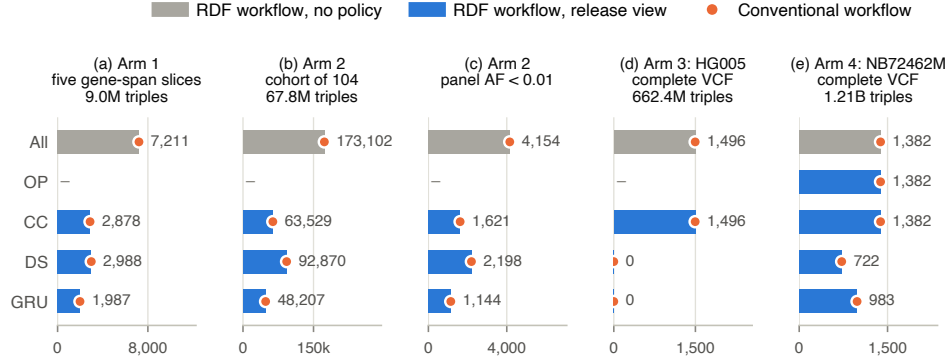


Fig. 4 The RDF and conventional workflows returned identical match sets for every requester in every arm. Counts are participant-variant-gene tuples, not people. Bars are the RDF workflow and dots the conventional workflow; they agree in every cell. CC denotes clinical care, DS disease-specific (cardiovascular) research, and GRU general research use; “All” applies no policy, and OP is care by the participant’s own physician, a requester only in Arm 4 (dashes elsewhere). Triple counts cover only each arm’s converted VCFs, without ClinVar or links. (c) joins the cohort’s matches to the separate panel-frequency graph at panel allele frequency (AF) < 0.01.

The cohort rare-variant query⁴ illustrates the benefit of representing shared information separately. In the 1000 Genomes inputs, each participant file repeats panel-level INFO fields describing all 3,202 panel samples. These fields accounted for 93% of the RDF generated per participant, or approximately 857 million triples across the cohort. The cohort workflow removes this repeated information and instead represents the required population frequencies once in a separate graph containing 68,635 sites and 6.5 million triples. SPDI links then connect each genomic observation to ClinVar and the shared frequency graph, while the conventional workflow retrieves the same frequencies by position and allele. Both routes returned the same rare-variant matches. The reduction therefore results from storing shared information once rather than from RDF compression.

Shared variant identity also changed how one release rule could be expressed. Restricting HFE C282Y to clinical care required only a policy change in the RDF

⁴Query file: [cohort/rare.rq](#).

workflow, without modifying the policy engine. The conventional workflow required corresponding changes to both code and configuration data. Because the RDF rule targeted the shared variant identity, it also selected the HG002 record represented on contig 6 rather than chr6. Three other rule changes required similarly small configuration changes in both workflows. These results demonstrate reusable policy configuration for this tested case, rather than a general advantage in usability or implementation effort. Supplementary Section S4 reports the complete change-impact and linking results.

Most of the linked workflow’s computational cost occurred before query execution (Figure 5). For Arm 3, conversion of the complete HG005 VCF (3,887,810 records after normalization) and ClinVar required 54 min and linking required 8 min. Materializing, validating, and indexing the clinical requester’s view required a further 2.5 h, including 68 min for validation. Peak resource use was 13.1 GB of memory and 34 GB of disk. Once these steps were complete, individual queries required 205–209 s. Query times across all four arms ranged from 146–209 s, suggesting that the fixed ClinVar join dominated this workload rather than that SPARQL query time was independent of graph size. In Arm 3, the clinical view contained every record of the HG005 VCF while the other two views were empty. In Arm 4, materializing and validating each selective view of the 1.2-billion-triple NB72462M graph required 55–74 min.

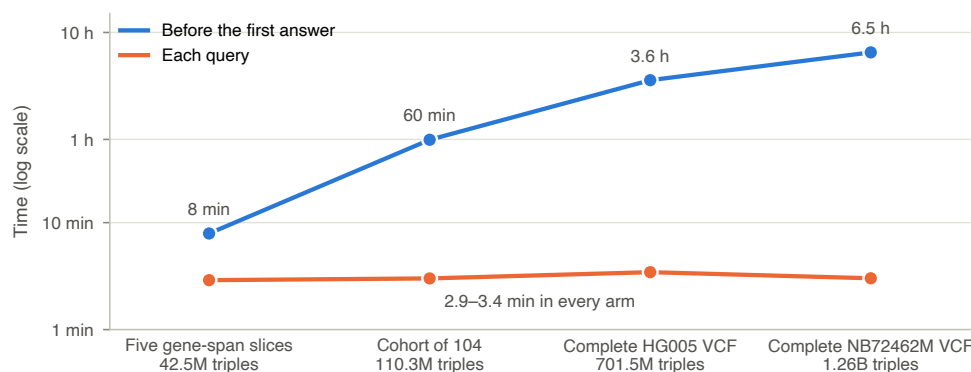


Fig. 5 Setup grows with the data; querying does not. Time before the clinical-care requester’s first answer (converting and linking the arm’s files, then writing, checking, and indexing its view) and the median time of each query, for the four arms in order of graph size. The size under each arm is the number of triples QLever indexed for that arm’s unrestricted (no-policy) query: the arm’s converted VCFs, the converted ClinVar subset (33.1M triples), Arm 2’s frequency graph, and all links. These sizes are therefore larger than the VCF-only triple counts in Figure 4. Supplementary Table S9 gives every stage for every requester.

4.3 RQ3: Representation and workload determine the practical costs

4.3.1 Sample granularity determines the amount of graph

The record ladder showed approximately linear growth in RDF output, producing 170.5–171.9 triples per VCF record. Mapping-stage resident memory remained between 1.0 and 1.7 GB as the number of records increased, while wall time and disk use increased with dataset size. These results indicate bounded memory use during mapping; they do not imply bounded memory requirements for subsequent validation, indexing, or querying.

The sample ladder revealed a substantially larger effect from the choice of sample representation (Figure 6). For 10,000 records and 2,504 samples, the expanded profile produced 627.4 million triples and a 4.10 GB HDT, compared with a 59.2 MB HDT for the condensed profile. The expanded representation contained $432\times$ more triples, although its HDT representation was only $69\times$ larger because compression reduced the difference in stored size. Across the sample ladder, the condensed representation’s triple count increased by only 0.9%, while HDT size increased $5.8\times$ as the encoded sample vectors became longer. This reduction in graph size comes at the cost of direct addressability: applications using the condensed profile must decode sample vectors rather than query individual sample calls as separate RDF resources. The complete 2,504-sample chromosome-20 VCF was not converted under either profile.

4.3.2 Separating core conversion from optional artifact construction

Space-optimized processing reduced transient workspace by $9.19\times$ for 100,000 HG005 records, with a 5.2% increase in runtime and no change to the generated triples. Across the corpus, COTTAS required 0.37–0.55 times the storage of gzip-framed N-Triples and 0.27–0.37 times that of HDT, but required 1.5–3.0 times longer than HDT to build. These differences reflect optional storage formats and their implementations rather than costs inherent to semantic conversion.

For the complete HG005 VCF, conversion required 16.07 h when both optional indexed formats were also generated. HDT construction accounted for 20,911 s and COTTAS construction for 31,230 s, together contributing approximately 14.5 h of the total runtime. Retaining HDT, its index, and COTTAS required 10.88 GB, whereas the gzip-framed N-Triples artifact alone required 2.89 GB. These values therefore represent alternative retained artifacts rather than a minimum storage requirement. The 16.07-hour measurement is also distinct from the later 54-minute use-case conversion, which used a different configuration and included ClinVar processing. Supplementary Section S8 reports the per-input artifact sizes, build times, and workspace measurements.

4.3.3 Indexed retrieval benefits some questions, not every workflow

On the 17.1-million-triple graph generated from 100,000 HG005 records, individual indexed QLever queries took 0.005–4.62 s, compared with 11.5–12.3 s for full cyvcf2

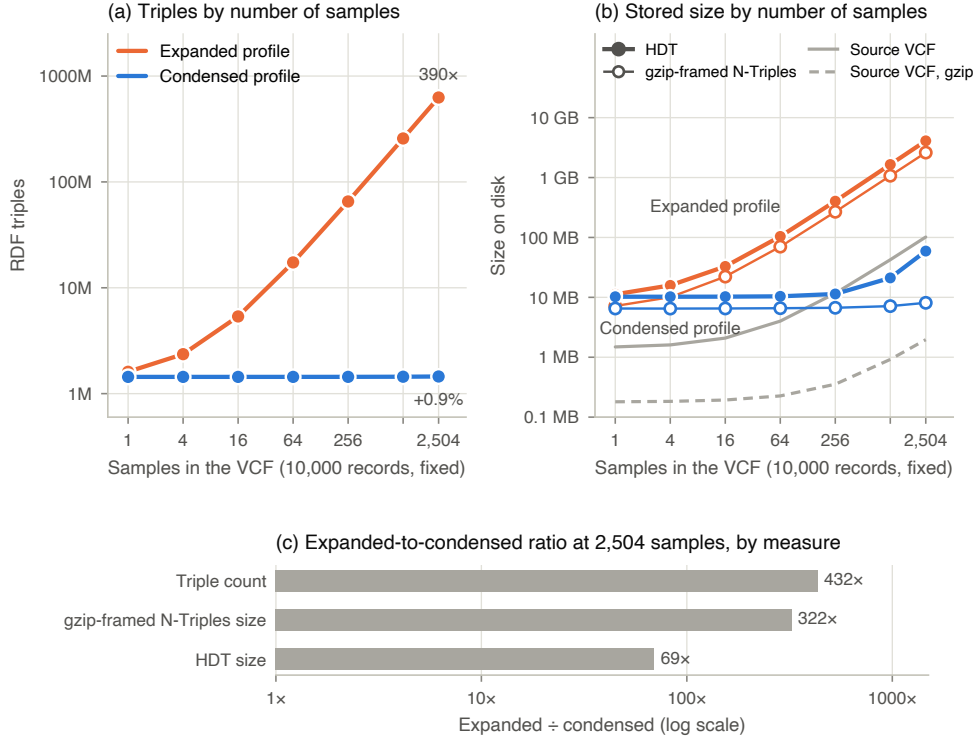


Fig. 6 Sample representation trades addressability for size. The same 10,000 records of the 1000 Genomes chromosome-20 VCF are converted with one to 2,504 of its samples, under each sample profile. (a) Expanded resources grow with sample calls; condensed vectors keep triple counts nearly constant. (b) Both profiles grow in stored size; the source VCF is shown uncompressed and gzip-compressed for reference. (c) At 2,504 samples the expanded-to-condensed ratio depends on the measure. The 390 $\times$ and +0.9% annotations in (a) instead compare each profile’s triple count at the two ends of the ladder. Medians are shown where replicated.

parses of the VCF (Figure 7). The QLever speed advantage applies only after the RDF graph has been converted and indexed; QLever indexing alone required 22.8s. For the complete thirteen-question workload, cyvcf2 answered all questions in a single 12.4s parsing pass, whereas the thirteen QLever queries required 17.1s even before setup costs were included. Parsing the VCF was therefore faster for a cold, one-off workload and for questions that could be combined into a single pass, while the existing QLever index was faster for selected queries executed repeatedly. In a separate three-replicate rerun that converted the records only to the gzip-framed N-Triples that QLever indexes, without the optional HDT, conversion took 95.7s and QLever indexing 25.0s, so setup required approximately 2.0 min. Under the tested workload, this initial cost was recovered after approximately 11 separately executed queries (Figure 8a). The gzip-framed N-Triples were 24.7 times the size of the gzipped VCF for these records, and 20.7 times for the complete HG005 VCF (Figure 8b).

Regional retrieval provides a stronger comparison, against region queries on a tabix-indexed VCF (Figure 8c). Across the 100,000-record HG005 graph and the

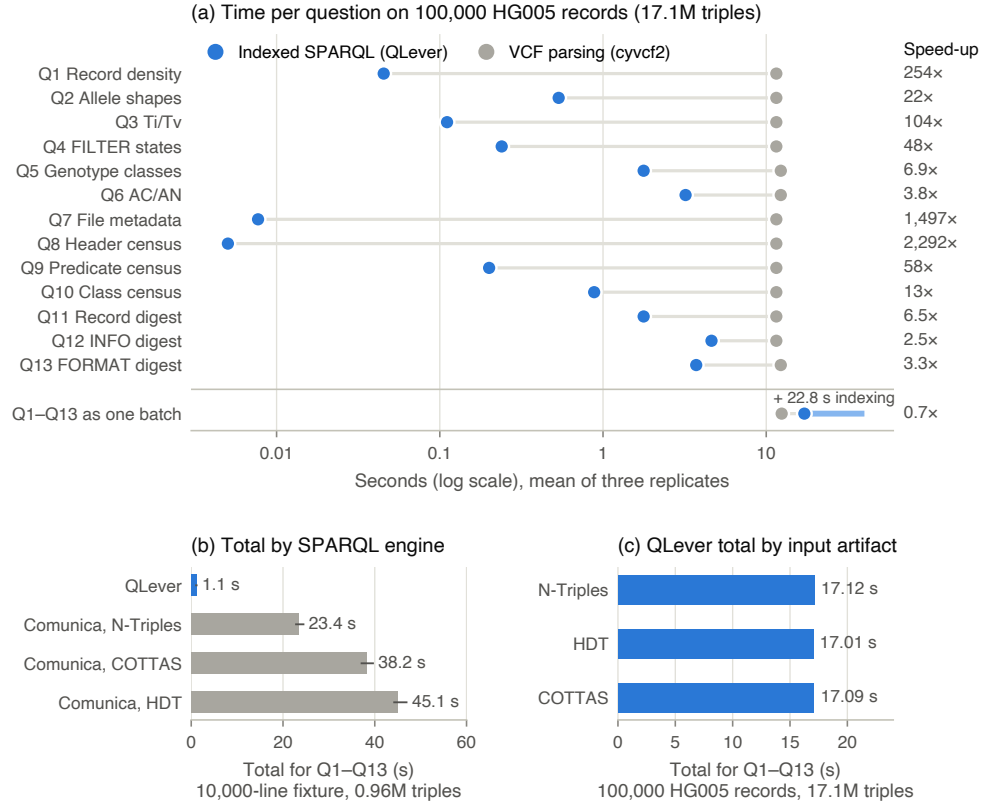


Fig. 7 Retrieval cost depends on the question and execution path. Answers agree before timings are compared. (a) Individual questions and the batch on the 17.1M-triple graph of 100,000 HG005 records, mean of three replicates; conversion is excluded and QLever indexing appears only on the batch row. Speed-up is the cyvcf2 parsing time divided by the SPARQL query time. (b) Four engines on the 0.96M-triple graph of the 10,000-line fixture, excluding setup; means and ranges over nine runs. Comunica queries N-Triples and HDT directly and COTTAS through its DuckDB-backed engine. (c) QLever on the 100,000-record graph loaded from three artifacts. Similar times in (c) describe QLever’s index, not native querying of those formats.

10,000-line fixture, 11,160 query executions completed without failures or answer disagreements. For 1 kb windows, tabix-indexed region queries with bcftools and cyvcf2 were faster than QLever. For the tested 10 Mb windows, QLever was faster. QLever query time remained nearly constant across window sizes, whereas tabix-indexed region queries became slower as more records were parsed. Setup costs differed substantially: QLever indexing required 23.2s, compared with 0.48s for bgzip and tabix.

Combining the regional and whole-file experiments clarifies when each approach was faster (Figure 8d). RDF querying outperformed VCF parsing for every region query that parsed the unindexed VCF, for 10 Mb windows queried through the tabix index with either tool, for 1 Mb windows with cyvcf2, and for whole-file questions executed individually. VCF parsing remained faster for tabix-indexed region queries up to 100 kb, for 1 Mb windows queried with bcftools, and for the single parsing pass that answered all thirteen whole-file questions.

On the 657.4M-triple graph of the complete HG005 VCF, QLever answered all thirteen inspection questions in 634.6s after an 868.0-second index build, with every answer equal to the parser's. This separate retrieval run compared answers without SHACL. Loading the same graph from HDT or COTTAS changed subsequent query time by only 0.3%, but decoding costs differed. The native HDT- and COTTAS-backed paths did not complete on the 171M-triple graph of one million HG005 records, despite answering five questions. Supplementary Table S14 reports both the completions and failures; small-fixture engine rankings do not establish large-graph feasibility.

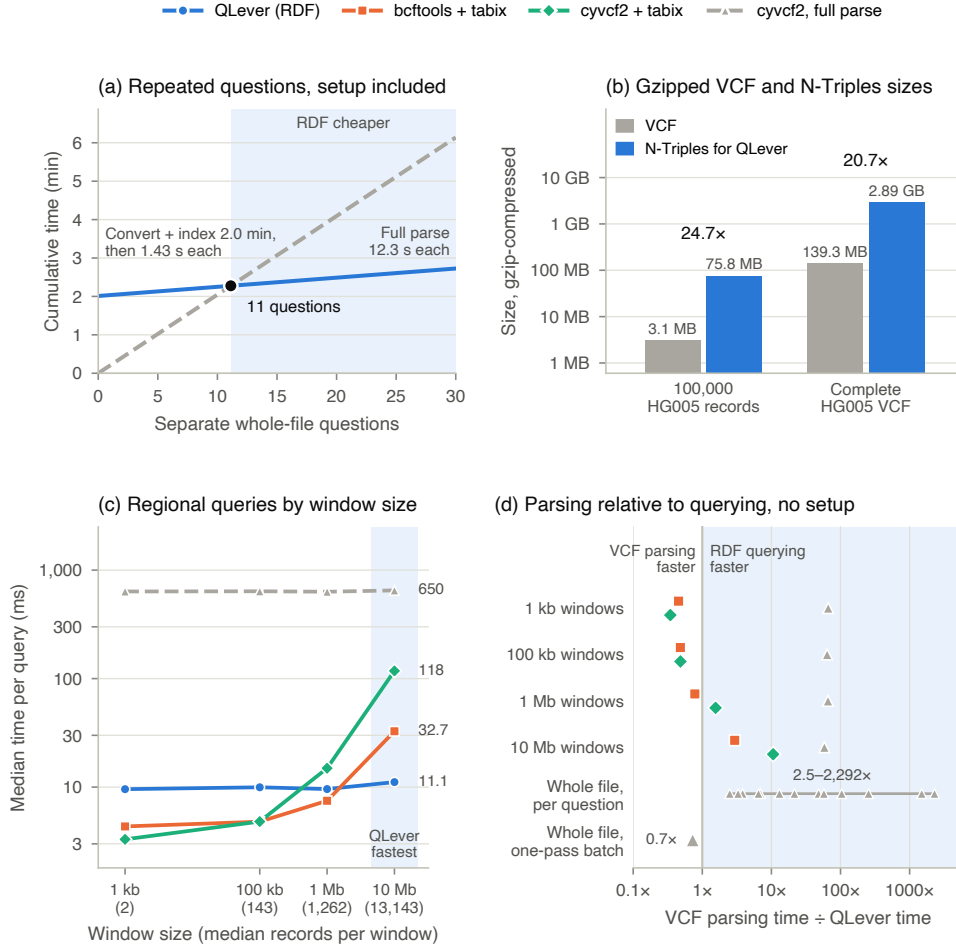


Fig. 8 Tabix-indexed VCF parsing is faster for small regions, and a single parsing pass for batched questions, while QLever is faster for large regions and, after a two-minute setup, for repeated queries. All panels use 100,000 HG005 records, represented as a 17.1-million-triple graph for QLever and as VCF for bcftools and cyvcf2, and (b) adds the complete HG005 VCF; all compared answers agreed. (a) Cumulative time for separately executed whole-file questions, including 2.0 min to convert the records to gzip-framed N-Triples and build the QLever index, measured in a three-replicate rerun; the cumulative times cross after 11 questions. (b) Gzip-compressed size of the VCF and of the N-Triples that QLever indexes. (c) Median regional-query time across five questions, twenty windows per size (three when parsing the unindexed VCF), and three replicates; parentheses report the median records per window. (d) Ratio of VCF parsing time to QLever query time for the regional and whole-file comparisons, excluding setup; values above one indicate faster QLever execution. Regional indexing required 23.2 s for QLever and 0.48 s for bgzip and tabix. Supplementary Figure S5 provides per-query results and break-even estimates, and Supplementary Table S15 reports the regional medians.

5 Discussion

5.1 A first step toward reusable genomic information

The three motivating use cases—clinical variant interpretation and reanalysis, pharmacogenomic clinical decision support, and consent-aware genomic data reuse—require more than converting VCF into another serialization. Each use case depends on explicit links between genomic observations and information maintained outside the source file, including external evidence, clinical context, and conditions governing data use. VCF-RDFizer provides an evaluated representation layer for expressing those links by preserving source context, validating the generated semantic representation, and demonstrating how integration with external resources and policy-aware release could work. The present study does not implement any of the three applications in full. The reported results therefore evaluate a reusable representation and workflow foundation rather than the clinical effectiveness, operational utility, or security of the applications that could be built on top of it.

The agreement with the bcftools workflow also shows that the evaluated integration task is not unique to RDF. The difference is representational: the RDF workflow expresses observations, external resources, shared identifiers, and policies within a common graph model that can be queried and extended without modifying the source representation. In the evaluated workflow, this allowed population frequencies to be stored once in a shared graph and a variant-specific release rule to be changed without modifying the policy engine. These results demonstrate reuse of information and rules within the tested setting, but do not establish a general reduction in implementation effort or imply that conventional pipelines cannot provide equivalent functionality. The following sections consider how this representation could support clinical reanalysis, pharmacogenomic decision support, and consent-aware reuse, and what additional components each application would require.

5.2 Clinical interpretation and reanalysis: keeping observations separate from changing evidence

For clinical variant interpretation and reanalysis, RDF could make the evidence supporting an interpretation more explicit and easier to inspect. A genomic observation could remain linked to the evidence used at a specific time, while later assertions are added with separate provenance rather than replacing the earlier interpretation. Queries could then identify observations affected by a revised clinical assertion or population-frequency estimate and flag the corresponding cases for review. Such a workflow could support the reevaluation process described in clinical guidance, where new knowledge or clinical information may alter a previous conclusion [8, 9]. RDF-based reanalysis would not replace clinical interpretation or recover variants absent from the original call set.

The ClinVar and population-frequency joins evaluated here demonstrate the static relationships required for such a workflow, but not longitudinal reanalysis. A practical implementation would first require explicit rules for reference assemblies, allele normalization, and unresolved matches. The minimally trimmed SPDI expressions

used by VCF-RDFizer do not necessarily correspond to NCBI contextual or canonical alleles in repetitive regions [26]. Biologically equivalent alleles can therefore receive different identifiers and fail to join with relevant evidence or identity-targeted policies. A reanalysis system would need normalization validated across representative allele classes and explicit handling of ambiguous or unmatched observations.

Longitudinal reanalysis would also require versioned evidence at the assertion level, including submitter, disease context, review status, supporting evidence, and relevant dates. Patient phenotypes, family information, and previous reports would require compatible representations linked to the correct sample and genomic observation. A change-detection process could then identify observations affected by updated evidence, refresh relevant queries or indexes, and route candidate changes for clinical review. Evaluation should compare flagged cases with conventional reannotation and expert-reviewed outcomes, including both missed updates and unnecessary reviews. Previous studies demonstrate the potential clinical value of genomic reanalysis, but that benefit cannot be inferred from the static query agreement measured in the present study [10].

5.3 Pharmacogenomic decision support: connecting the genomic observation to the clinical question

For pharmacogenomic clinical decision support, RDF could represent the path from a genomic observation to an interpreted pharmacogenomic result, a medication order, and the guidance relevant to prescribing. Separate linked resources could preserve which genomic observation, interpretation, and guideline version support a recommendation and could allow a previously derived interpretation to be reused when prescribing context changes. Such a model addresses an integration challenge identified in pharmacogenomic decision-support research without replacing established clinical systems or interpretation tools [12, 13].

The present study does not implement pharmacogenomic interpretation. Gene overlap and ClinVar classification are insufficient to infer a pharmacogenomic diplo-type, metabolizer phenotype, or treatment recommendation. A practical implementation would first connect represented variant calls to a validated pharmacogenomic interpretation pipeline, such as an appropriately configured PharmCAT workflow, while retaining the pipeline assumptions, supporting evidence, and unresolved results [2]. The genomic input would also need adequate coverage, phasing, and representation of clinically relevant variants. Missing VCF records cannot automatically be interpreted as reference genotypes or negative findings. Additional calling or interpretation methods would therefore be required when the source data do not satisfy the requirements of the pharmacogenomic model. RDF could preserve the resulting interpretations and provenance, but the biological inference would remain the responsibility of the interpretation pipeline.

Clinical deployment would then require reliable links between the interpreted pharmacogenomic result, patient, medication, indication, and other relevant clinical context using agreed identifiers and exchange models, including FHIR Genomics where appropriate [41]. Computable prescribing guidance would require explicit versioning, rule representation, and handling of missing or conflicting information. The

CYP2C19–clopidogrel example illustrates the need for that additional context because genotype-derived phenotype alone does not determine the prescribing recommendation [11]. A clinical rule engine and EHR integration would then be needed to deliver the recommendation at the appropriate point in care. Evaluation would require expert validation of pharmacogenomic interpretations, comparison of generated recommendations with the source guidance, and assessment of the resulting clinical workflow and user interaction [12]. The present study contributes a validated genomic representation and an evaluated demonstration of cross-resource linking that could support such a workflow.

5.4 Consent-aware reuse: connecting explicit policies to an enforced release process

For the third use case, RDF could make the association between a genomic resource, its governing policy, and the purpose of a request directly inspectable. Policies could refer to identified files, participants’ records, or declared selections without requiring a separate copy of the source file for every permitted use. The evaluated plug-in demonstrates part of this behavior: requester-specific views agreed with the conventional route, and ownership validation detected context that should have been withheld with its source. The views differed by participant, gene panel, purpose, region, and single variant, and arm 4 released four different, non-empty parts of one complete VCF. The study does not establish masking within one multi-sample file, and region rules match contig names as each file writes them.

An operational system would first require the governing consent and institutional conditions to be reviewed and mapped to a defined policy profile. DUO and the GA4GH Machine Readable Consent Guidance support consistent descriptions, but a computable term does not establish that a particular use is ethically or legally authorized [7, 14, 15]. The profile would need agreed meanings for purposes, permissions, prohibitions, conflicts, and resource ownership, together with a process for resolving conditions that cannot be evaluated automatically. Default-deny and deny-overrides are choices in our implementation, not guarantees supplied by RDF or ODRL [27].

Policy evaluation must then be connected to authenticated requester credentials and an enforcement boundary that protects the source graphs, indexes, caches, and exports as well as the query endpoint [7, 28]. Policy versions, decisions, and released views would need auditable records. A withdrawal or policy change would require recomputing affected permissions and controlling future access; invalidating a local view would not by itself retract copies already released. Tests would also need to cover equivalent variant expressions, dependent resources, multi-sample representations, and information that might still be inferred from a permitted result. Security and privacy assessments are therefore separate from source-fidelity validation. The plug-in is a first step toward inspectable release decisions, not anonymization, a guarantee of regulatory compliance, or protection against access outside the evaluator. Obligations are recorded but not enforced in the evaluated implementation.

5.5 Validation needs to test both the graph and its oracle

For consent-aware genomic data reuse, RDF could make the relationship between a genomic resource, the policy governing access, and the purpose of a request directly inspectable. Policies could target identified files, participant records, or declared subsets without requiring a separate source copy for every permitted use. The evaluated policy plug-in demonstrates part of this model: requester-specific release views matched the conventional workflow, and ownership validation identified dependent context that also required removal. Across the evaluated scenarios, release decisions varied by participant, purpose, gene panel, genomic region, and individual variant. Arm 4 produced four distinct, non-empty release views from the same complete VCF. The current evaluation does not test selective masking within a multi-sample VCF, and region-based rules still depend on the contig labels present in each source file.

Operational deployment would first require explicit mapping from consent terms and institutional conditions to a defined policy profile. DUO and the GA4GH Machine Readable Consent Guidance provide machine-readable terminology for describing permitted uses, but machine-readable representation alone does not determine whether a specific request is ethically or legally authorized [7, 14, 15]. A production profile would therefore need agreed semantics for purposes, permissions, prohibitions, conflict resolution, and resource ownership, together with procedures for conditions that cannot be evaluated automatically. The default-deny and deny-overrides semantics used in VCF-RDFizer are implementation choices rather than properties guaranteed by RDF or ODRL [27].

Policy evaluation would also need to operate within an authenticated and enforced access-control architecture. The enforcement boundary would need to protect source graphs, indexes, caches, exported views, and query interfaces, rather than only the policy evaluator [7, 28]. Policy versions, access decisions, and released views would require auditable records. Withdrawals and policy changes would require reevaluation of future access, while previously exported copies could not be revoked solely by updating the local policy state. Evaluation would also need to cover equivalent variant representations, dependent resources, multi-sample data, and information that could still be inferred from permitted outputs.

Security, privacy, and regulatory compliance therefore remain distinct from source-fidelity and policy-evaluation tests. The evaluated plug-in demonstrates inspectable, requester-specific release decisions under simulated rules, but does not provide anonymization, regulatory assurance, or protection against access outside the enforcement boundary. The current implementation records obligations but does not enforce them.

5.6 Representation and workload determine the deployment choice

Semantic representation introduces computational and storage costs, and the appropriate representation depends on the intended workload. The expanded sample profile preserves individual sample calls as directly addressable RDF resources, whereas the

condensed profile reduces graph size by encoding sample values in vectors. The condensed representation therefore lowers storage costs at the expense of direct querying, validation, and policy targeting of individual sample values. For example, sample-level validation requires vector-aware logic, and withholding a single value would require modifying the corresponding vector. Triple count alone consequently does not capture the practical cost or functionality of either representation.

Parsing the VCF directly remains preferable for many one-pass analyses and small region queries on a tabix-indexed VCF because no semantic conversion or graph indexing is required. The additional RDF setup becomes more defensible when the same data support repeated queries, links to external knowledge, or policy-dependent access. For example, a clinical reanalysis system could retain a converted genome while repeatedly querying updated evidence, rather than reconstructing the required relationships for every reanalysis cycle. The present experiments demonstrate the potential for such reuse, but do not measure a longitudinal reanalysis workload.

The measured 11-query break-even point illustrates this trade-off rather than defining a general threshold (Figure 8a). The estimate includes 95.7 s to convert the records to gzip-framed N-Triples plus 25.0 s for QLever indexing, offset by approximately 10.9 s saved for each separately executed question. The estimate therefore depends on the tested query mix and on a full parse of the VCF for every individual question, and a single parsing pass that answers all questions changes the comparison (Supplementary Section S8). The graph route also needs more storage: the gzip-framed N-Triples that QLever indexes were 21–25 times the size of the gzipped VCF (Figure 8b). Optional formats such as HDT and COTTAS should likewise be generated only when their storage or access properties are required; gzip-framed N-Triples with an appropriate index was sufficient for the evaluated QLever workflow. Applications based on changing evidence or access policies would additionally incur costs for refreshing links, indexes, or release views, which were not benchmarked here.

5.7 Scope and next steps

The present results support VCF-RDFizer as a software and workflow contribution rather than evidence of clinical deployment, measured usability, or superiority over other RDF converters. The converter comparison is based on published documentation, while the workflow comparison uses bcftools as the conventional reference. The sample-scaling experiment holds INFO content fixed, the complete multi-sample cohort was not converted, and the most comprehensive SHACL profiles were evaluated only on small inputs. The linking and policy experiments use single-sample files and simulated access rules, so all reported outcomes remain specific to the tested datasets, software versions, reference releases, and policy configurations.

Future work can therefore proceed in stages. The first priority is to strengthen the representation and validation layer by closing remaining oracle-coverage gaps, extending conformant sites-only support, and defining clearer boundaries for variant normalization and shared identity. The second stage is application-specific evaluation. Clinical reanalysis would require versioned evidence updates and change detection; pharmacogenomic decision support would require validated interpretation linked to clinical context; and consent-aware reuse would require selective release

under reviewed policy models. Each application should be evaluated with the relevant domain experts and against application-specific reference workflows.

A later deployment stage would require the clinical, governance, privacy, and security assessments appropriate to the intended environment. Comparative claims could also be strengthened through shared-input evaluation against other VCF-to-RDF converters. GA4GH VRS identifiers, federated querying, and policy evaluation over multi-sample data remain promising extensions but are not demonstrated by the present study.

VCF-RDFizer therefore provides a testable foundation for this progression: a genomic representation that can be validated against its source, linked to external information, and reused across workflows while the additional requirements of each application remain explicit.

6 Conclusions

VCF-RDFizer makes VCF information explicit in a representation that can be checked against its source, connected to external knowledge, and used in policy-aware research queries. The evaluation demonstrates complementary validation, reusable joins, and requester-specific result agreement with a conventional workflow, while exposing limits in oracle coverage, identity assumptions, and large-graph validation. Its measured costs support choosing semantic conversion for tasks that need those relationships and validation, rather than treating RDF as a universal replacement for native VCF processing. This is a first step toward evidence-aware reanalysis, pharmacogenomic decision support, and governed reuse; realizing those applications still requires validated interpretation, clinical integration, and enforced data-access processes.

Availability and requirements

Project: VCF-RDFizer. **Home page:** <https://github.com/ecrum19/VCF-RDFizer>. **Language:** Python, with the mapping, compression, and query tools specified in Supplementary Section S5. **Platforms:** Linux, macOS, and Windows are covered by the tool’s tests; the experiments reported here ran on Ubuntu. **Distribution:** PyPI, Conda-Forge, and Docker. **License:** MIT. **Any restrictions to use by non-academics:** none. Docker supplies the pinned benchmark environment; requirements for other installations are documented with each release. v3.1.0 produced the base benchmark, v3.2.0 converted the use-case inputs, v3.3.0 releases the linking and policy extensions evaluated in pre-release builds, and v3.3.1 provided the linking for Arm 4 and the validator for the consumer WGS validation run. The release ledger distinguishes those builds rather than attributing every result to one version, although v3.3.1 should be able to reproduce all data generated in this study.

Declarations

Ethics approval and consent to participate

This study used publicly released genomic data and involved no recruitment of or interaction with participants. Personal Genome Project donors released their files under the project’s open-consent model [73]; GIAB and precisionFDA files are public reference materials [74, 77]; the 1000 Genomes and HGSVC2 datasets are released for research reuse [75, 76, 79]; and ClinVar is a public NCBI resource [80]. The experimental consents are simulated and do not express any real participant’s wishes. No identification of participants was attempted. Ethics approval was not sought for the reported work.

Consent for publication

Not applicable. The manuscript reports aggregate results and software measurements, not individual-level genomic records, images, or personal details.

Availability of data and materials

Input filenames, provider URLs, download commands, fixtures, derivation scripts, query files, manifests, normalized outputs, and comparison reports are maintained at <https://github.com/ecrum19/vcf-rdfizer-testing>. Large source VCFs are obtained from their public providers rather than redistributed in Git. Experiment 17 (`benchmarks/use_case/acmg`) contains the use-case definitions, simulated policies, conventional baseline, and applied-change tests. Its match sets, timings, and check reports, as well as the additional validation runs, are recorded under `BioMedSem_2026/benchmark-results`. No biological materials were generated or used.

The software is available at the project home page. The base benchmark used v3.1.0 (`d3b34d5`), and the linking and policy extensions are released in v3.3.0 (`7ab2300`; image `sha256:1838720d...06432`). The use case ran pre-release builds, including `578c2be` for Arm 3, rather than the exact final v3.3.0 release. Its tested inputs did not meet the empty-selector or no-VCF-triples conditions subsequently refused by the plug-ins. Supplementary Section S5 identifies the pre-release and later-run commits, the v3.1.0 image digest, and the bundled shape snapshot. The public vocabulary is separately versioned [51]. Results must be traced to their recorded build and input, not only to the latest package.

[AUTHOR QUERY: Add a persistent archive identifier containing the exact evaluated code, dirty-tree changes for use-case arms 1–2, complete image digests, vocabulary/shapes, queries, and evidence manifests. Include the commits and configurations of the default-shape mutation rerun and the consumer WGS validation runs recorded in the release ledger. The supplied manuscript package does not itself establish this complete archived snapshot.]

Competing interests

None declared.

Funding

Elias Crum acknowledges support from the Research Foundation–Flanders (FWO), SB Fellowship 1S27825N.

[AUTHOR QUERY: Complete the funding statements for B.B., R.T., and G.E.]

Authors' contributions

E.C., G.E., and R.T. conceived of this study. E.C. did the implementation and coding work. E.C. devised the validation and use-case demonstration with the guidance of R.T. and executed the experiments. E.C., G.E., and R.T. reviewed and verified all results. E.C., B.B., G.E., and R.T. wrote and edited the manuscript.

Acknowledgments

Benchmarking used the resources of IDLab at Ghent University. Claude Opus 5.5 [81] and GPT-6 Astra [82] assisted with experiment automation scripting, result data analysis scripting, and restructuring and language editing of this manuscript. The authors reviewed all AI outputs and take full responsibility for the manuscript, experiments, tools, and repositories.

References

- [1] SAM/BAM Format Specification Working Group: The Variant Call Format Specification (2024). <https://github.com/samtools/hts-specs>
- [2] Li, B., Sangkuhl, K., Keat, K., Whaley, R.M., Woon, M., Verma, S., Dudek, S., Tuteja, S., Verma, A., Whirl-Carrillo, M., Ritchie, M.D., Klein, T.E.: How to run the pharmacogenomics clinical annotation tool (pharmcat). *Clinical Pharmacology & Therapeutics* **113**(5), 1036–1047 (2023) <https://doi.org/10.1002/cpt.2790>
- [3] Souche, E., Beltran, S., Brosens, E., Belmont, J.W., Fossum, M., Riess, O., Gilissen, C., Ardeshirdavani, A., Houge, G., Van Gijn, M., Clayton-Smith, J., Synofzik, M., De Leeuw, N., Deans, Z.C., Dincer, Y., Eck, S.H., Van Der Crabben, S., Balasubramanian, M., Graessner, H., Sturm, M., Firth, H., Ferlini, A., Nabhout, R., De Baere, E., Liehr, T., Macek, M., Matthijs, G., Scheffer, H., Bauer, P., Yntema, H.G., Weiss, M.M.: Recommendations for whole genome sequencing in diagnostics for rare diseases. *European Journal of Human Genetics* **30**(9), 1017–1021 (2022) <https://doi.org/10.1038/s41431-022-01113-x>
- [4] Danecek, P., Bonfield, J.K., Liddle, J., Marshall, J., Ohan, V., Pollard, M.O., Whitwham, A., Keane, T., McCarthy, S.A., Davies, R.M., Li, H.: Twelve years of samtools and bcftools. *GigaScience* **10**(2), 008 (2021) <https://doi.org/10.1093/gigascience/giab008>
- [5] Gottesman, O., Kuivaniemi, H., Tromp, G., Faucett, W.A., Li, R., Manolio, T.A., *et al.*: The Electronic Medical Records and Genomics (eMERGE) Network: past, present, and future. *Genetics in Medicine* **15**(10), 761–771 (2013) <https://doi.org/10.1038/gim.2013.72>
- [6] The All of Us Research Program Genomics Investigators: Genomic data in the All of Us research program. *Nature* **627**(8003), 340–346 (2024) <https://doi.org/10.1038/s41586-023-06957-x>
- [7] Rehm, H.L., *et al.*: GA4GH: International policies and standards for data sharing across genomic research and healthcare. *Cell Genomics* **1**(2), 100029 (2021) <https://doi.org/10.1016/j.xgen.2021.100029>
- [8] Richards, S., Aziz, N., Bale, S., Bick, D., Das, S., *et al.*: Standards and guidelines for the interpretation of sequence variants: a joint consensus recommendation of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology. *Genetics in Medicine* **17**(5), 405–424 (2015) <https://doi.org/10.1038/gim.2015.30>
- [9] Deignan, J.L., Chung, W.K., Kearney, H.M., Monaghan, K.G., Rehder, C.W., Chao, E.C.: Points to consider in the reevaluation and reanalysis of genomic test results: a statement of the American College of Medical Genetics and Genomics

- (ACMG). *Genetics in Medicine* **21**(6), 1267–1270 (2019) <https://doi.org/10.1038/s41436-019-0478-1>
- [10] Hiatt, S.M., Amaral, M.D., Bowling, K.M., Finnila, C.R., Thompson, M.L., Gray, D.E., Lawlor, J.M.J., Cochran, J.N., Bebin, E.M., Brothers, K.B., East, K.M., Kelley, W.V., Lamb, N.E., Levy, S.E., Lose, E.J., Neu, M.B., Rich, C.A., Simmons, S., Myers, R.M., Barsh, G.S., Cooper, G.M.: Systematic reanalysis of genomic data improves quality of variant interpretation. *Clinical Genetics* **94**(1), 174–178 (2018) <https://doi.org/10.1111/cge.13259>
 - [11] Lee, C.R., Luzum, J.A., Sangkuhl, K., Gammal, R.S., Sabatine, M.S., Stein, C.M., Kisor, D.F., Limdi, N.A., Lee, Y.M., Scott, S.A., Hulot, J.-S., Roden, D.M., Gaedigk, A., Caudle, K.E., Klein, T.E., Johnson, J.A., Shuldiner, A.R.: Clinical pharmacogenetics implementation consortium guideline for CYP2C19 genotype and clopidogrel therapy: 2022 update. *Clinical Pharmacology & Therapeutics* **112**(5), 959–967 (2022) <https://doi.org/10.1002/cpt.2526>
 - [12] Wake, D.T., Smith, D.M., Kazi, S., Dunnenberger, H.M.: Pharmacogenomic clinical decision support: A review, how-to guide, and future vision. *Clinical Pharmacology & Therapeutics* **112**(1), 44–57 (2022) <https://doi.org/10.1002/cpt.2387>
 - [13] Johnson, D., Del Fiol, G., Kawamoto, K., Romagnoli, K.M., Sanders, N., Isaacson, G., Jenkins, E., Williams, M.S.: Genetically guided precision medicine clinical decision support tools: a systematic review. *Journal of the American Medical Informatics Association* **31**(5), 1183–1194 (2024) <https://doi.org/10.1093/jamia/ocae033>
 - [14] Lawson, J., Cabili, M.N., Kerry, G., *et al.*: The Data Use Ontology to streamline responsible access to human biomedical datasets. *Cell Genomics* **1**(2), 100028 (2021) <https://doi.org/10.1016/j.xgen.2021.100028>
 - [15] Global Alliance for Genomics and Health: Machine Readable Consent Guidance (MRCG). Technical implementation guidance, version 1.0. Accessed 7 October 2026. <https://www.ga4gh.org/product/machine-readable-consent-guidance/> Accessed 2026-10-07
 - [16] European Parliament and Council of the European Union: Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation). *Official Journal of the European Union*, L 119, 4 May 2016, pp. 1–88 (2016). <https://eur-lex.europa.eu/eli/reg/2016/679/oj>
 - [17] Shabani, M., Borry, P.: Rules for processing genetic data for research purposes in view of the new EU General Data Protection Regulation. *European Journal of Human Genetics* **26**(2), 149–156 (2018) <https://doi.org/10.1038/>

- [18] Paila, U., Chapman, B.A., Kirchner, R., Quinlan, A.R.: GEMINI: integrative exploration of genetic variation and genome annotations. *PLoS Computational Biology* **9**(7), 1003153 (2013) <https://doi.org/10.1371/journal.pcbi.1003153>
- [19] Cyganiak, R., Wood, D., Lanthaler, M.: RDF 1.1: Concepts and Abstract Syntax (2014). <https://www.w3.org/TR/2014/REC-rdf11-concepts-20140225/>
- [20] Hogan, A., Blomqvist, E., Cochez, M., D’Amato, C., Melo, G., Gutierrez, C., Kirrane, S., Labra Gayo, J.E., Navigli, R., Neumaier, S., Ngonga Ngomo, A.-C., Polleres, A., Rashid, S.M., Rula, A., Schmelzeisen, L., Sequeda, J., Staab, S., Zimmermann, A.: Knowledge graphs. *ACM Computing Surveys* **54**(4) (2021) <https://doi.org/10.1145/3447772>
- [21] World Wide Web Consortium (W3C): LinkedData. <https://www.w3.org/wiki/LinkedData> Accessed 2026-03-04
- [22] Belleau, F., Nolin, M.-A., Tourigny, N., Rigault, P., Morissette, J.: Bio2RDF: Towards a mashup to build bioinformatics knowledge systems. *Journal of Biomedical Informatics* **41**(5), 706–716 (2008) <https://doi.org/10.1016/j.jbi.2008.03.004>
- [23] Breit, A., Ott, S., Agibetov, A., Samwald, M.: OpenBioLink: A benchmarking framework for large-scale biomedical link prediction (2019) <https://doi.org/10.48550/ARXIV.1912.04616> . Publisher: arXiv Version Number: 2
- [24] SIB Swiss Institute of Bioinformatics RDF Group Members, Altenhoff, A., Bairoch, A., Bansal, P., Baratin, D., Bastian, F., Bolleman*, J., Bridge, A., Burdet, F., Cramer, K., Dauvillier, J., Dessimoz, C., Gehant, S., Glover, N., Gnodtke, K., Hayes, C., Ibberson, M., Kriventseva, E., Kuznetsov, D., Frédérique, L., Mehl, F., Mendes De Farias*, T., Michel, P.-A., Moretti, S., Morgat, A., Österle, S., Pagni, M., Redaschi, N., Robinson-Rechavi, M., Samarasinghe, K., Sima, A.-C., Szklarczyk, D., Topalov, O., Touré, V., Unni, D., Von Mering, C., Wollbrecht, J., Zahn-Zabal*, M., Zdobnov, E.: The SIB Swiss Institute of Bioinformatics Semantic Web of data. *Nucleic Acids Research* **52**(D1), 44–51 (2024) <https://doi.org/10.1093/nar/gkad902>
- [25] Harris, S., Seaborne, A., Prud’hommeaux, E.: SPARQL 1.1 Query Language (2013). <https://www.w3.org/TR/2013/REC-sparql11-query-20130321/>
- [26] Holmes, J.B., Moyer, E., Phan, L., Maglott, D., Kattman, B.: SPDI: data model for variants and applications at NCBI. *Bioinformatics* **36**(6), 1902–1907 (2020) <https://doi.org/10.1093/bioinformatics/btz856>
- [27] Iannella, R., Villata, S.: ODRL Information Model 2.2. W3C (2018). <https://www.w3.org/TR/odrl-model/> Accessed 2025-11-27

- [28] OASIS: eXtensible Access Control Markup Language (XACML) Version 3.0. OASIS Standard (2013). <https://docs.oasis-open.org/xacml/3.0/xacml-3.0-core-spec-os-en.html>
- [29] Katayama, T., Kawashima, S., Micklem, G., Kawano, S., Kim, J.-D., Kocbek, S., Okamoto, S., Wang, Y., Wu, H., Yamaguchi, A., *et al.*: Biohackathon series in 2013 and 2014: improvements of semantic interoperability in life science data and services. *F1000Research* **8**, 1677 (2019) <https://doi.org/10.12688/f1000research.18238.1>
- [30] Lindenbaum, P.: JVarkit: java-based utilities for Bioinformatics. figshare (2015). <https://doi.org/10.6084/m9.figshare.1425030.v1> . https://figshare.com/articles/journal_contribution/JVarkit_java_based_utilities_for_Bioinformatics/1425030
- [31] Garrison, E., Kronenberg, Z.N., Dawson, E.T., Pedersen, B.S., Prins, P.: A spectrum of free software tools for processing the VCF variant call format: vcflib, bio-vcf, cyvcf2, hts-nim and slivar. *PLOS Computational Biology* **18**(5), 1009123 (2022) <https://doi.org/10.1371/journal.pcbi.1009123>
- [32] Mitsuhashi, N., Toyo-oka, L., Katayama, T., Kawashima, M., Kawashima, S., Miyazaki, K., Takagi, T.: TogoVar: A comprehensive Japanese genetic variation database. *Human Genome Variation* **9**(1), 44 (2022) <https://doi.org/10.1038/s41439-022-00222-9>
- [33] Bodrug-Schepers, A., Chabane, H., Montoya, G., Serrano-Alvarado, P., Redon, R., Gaignard, A.: Semantic Beacons: a framework to support federated querying over genomic variants and public Knowledge Graphs. In: *Semantic Web Applications and Tools for Health Care and Life Sciences (SWAT4HCLS 2025)*. CEUR Workshop Proceedings, vol. 4196. CEUR Workshop Proceedings, Barcelona, Spain (2025). https://ceur-ws.org/Vol-4196/paper_16.pdf Accessed 2025-11-24
- [34] Böhmer, J.K., Janssen, R.: Linked data in practice: An RDF-based approach with SPARQLing Genomics. Zenodo. Zenodo record, version 4 (2020). <https://doi.org/10.5281/zenodo.3903111>
- [35] Janssen, R.: SPARQLing Genomics. Open-source software. Accessed 2026-09-03 (2019). <https://gitlab.com/roelj/sparqling-genomics>
- [36] Baran, J., Durgahee, B.S.B., Eilbeck, K., Antezana, E., Hoehndorf, R., Dumontier, M.: GFVO: the Genomic Feature and Variation Ontology. *PeerJ* **3**, 933 (2015) <https://doi.org/10.7717/peerj.933>
- [37] Hail Team: Hail: scalable genomic data analysis. Accessed October 2026 (2026). <https://github.com/hail-is/hail>
- [38] TileDB, Inc.: TileDB-VCF: efficient variant-call data storage and retrieval.

- Accessed October 2026 (2026). <https://github.com/TileDB-Inc/TileDB-VCF>
- [39] Rambla, J., Baudis, M., Ariosa, R., *et al.*: Beacon v2 and Beacon networks: A “lingua franca” for federated data discovery in biomedical genomics, and beyond. *Human Mutation* **43**(6), 791–799 (2022) <https://doi.org/10.1002/humu.24369>
 - [40] Kelleher, J., Lin, M., Albach, C.H., Birney, E., *et al.*: htsget: a protocol for securely streaming genomic data. *Bioinformatics* **35**(1), 119–121 (2019) <https://doi.org/10.1093/bioinformatics/bty492>
 - [41] HL7 International: HL7 FHIR Genomics Reporting Implementation Guide. Accessed October 2026 (2026). <https://hl7.org/fhir/uv/genomics-reporting/>
 - [42] Penha, E.D.S., Iriabho, E., Dussaq, A., Oliveira, D.M., Almeida, J.S.: Isomorphic semantic mapping of variant call format (VCF2RDF). *Bioinformatics* **33**(4), 547–548 (2017) <https://doi.org/10.1093/bioinformatics/btw652>
 - [43] Strozzi, F.: bioruby-vcf2rdf: simple converter from VCF format to RDF Turtle format. GitHub repository, BioHackathon 2013. Accessed 2026-10-07 (2013). <https://github.com/fstrozzi/bioruby-vcf2rdf>
 - [44] BioInterchange: BioInterchange 2: genomic linked data converter for GFF, GVF, and VCF via JSON-LD. GitHub repository, version 2.0.5. Accessed 2026-10-07 (2021). <https://github.com/BioInterchange/BioInterchangeC>
 - [45] Bendiken, A., Bonnal, R.J.P., Strozzi, F.: ruby-rdf/rdf-vcf: RDF VCF parser (2015). <https://github.com/ruby-rdf/rdf-vcf> Accessed 2025-03-01
 - [46] Prasanna, S., Kumar, A., Rao, D., Simoes, E.J., Rao, P.: A scalable tool for analyzing genomic variants of humans using knowledge graphs and graph machine learning. *Frontiers in Big Data* **7**, 1466391 (2025) <https://doi.org/10.3389/fdata.2024.1466391>
 - [47] Larmande, P., Pittolat, B., Tando, N., Pomie, Y., Happi Happi, B.G., Guignon, V., Ruiz, M.: AgroLD: a knowledge graph for the plant sciences. *BMC Genomic Data* **26**(Suppl 1), 73 (2025) <https://doi.org/10.1186/s12863-025-01359-6>
 - [48] Touré, V., Unni, D., Krauss, P., Abdelwahed, A., Buchhorn, J., Hinderling, L., Geiger, T.R., Österle, S.: The SPHN Schema Forge – transform healthcare semantics from human-readable to machine-readable by leveraging semantic web technologies. *Journal of Biomedical Semantics* **16**, 9 (2025) <https://doi.org/10.1186/s13326-025-00330-9>
 - [49] Gaignard, A., Rosnet, T., Lamotte, F., Lefort, V., Devignes, M.-D.: FAIR-Checker: supporting digital resource findability and reuse with knowledge graphs and semantic web standards. *Journal of Biomedical Semantics* **14**, 7 (2023) <https://doi.org/10.1186/s13326-023-00289-5>

- [50] Ansari, A., Conte, M., Flynn, A., Paturkar, A.: A prototype ETL pipeline that uses HL7 FHIR RDF resources when deploying pure functions to enrich knowledge graph patient data. *Journal of Biomedical Semantics* **16**, 16 (2025) <https://doi.org/10.1186/s13326-025-00335-4>
- [51] Crum, E.: VCF Core Vocabulary. Version 2.1.3. GitHub repository. Version 2.1.3 with the SSSOM alignment set; source snapshot of 22 September 2026 (2026). <https://github.com/ecrum19/vcf-core-vocabulary/tree/8a5f23a>
- [52] Crum, E., Buelens, B., Ertaylan, G., Taelman, R.: VCF Core Vocabulary: A Semantic Foundation for RDF Representations of VCF. Manuscript in preparation for SWAT4HCLS 2027; preprint at commit 9507dd9 (2026). https://github.com/ecrum19/SWAT4HCLS_2027/blob/9507dd9/long_revised/main_long.pdf
- [53] Landrum, M.J., Lee, J.M., Benson, M., Brown, G.R., Chao, C., Chitipiralla, S., Gu, B., Hart, J., Hoffman, D., Jang, W., Karapetyan, K., Katz, K., Liu, C., Maddipatla, Z., Malheiro, A., McDaniel, K., Ovetsky, M., Riley, G., Zhou, G., Holmes, J.B., Kattman, B.L., Maglott, D.R.: ClinVar: improving access to variant interpretations and supporting evidence. *Nucleic Acids Research* **46**(D1), 1062–1067 (2018) <https://doi.org/10.1093/nar/gkx1153>
- [54] VCF-RDFizer maintainers: vcf-rdfizer. Python Package Index (PyPI), version 3.3.0. Accessed 2026-10-07 (2026). <https://pypi.org/project/vcf-rdfizer/>
- [55] VCF-RDFizer maintainers: vcf-rdfizer. conda-forge, version 3.3.0. Accessed 2026-10-07 (2026). <https://anaconda.org/conda-forge/vcf-rdfizer>
- [56] VCF-RDFizer maintainers: ecrum19/vcf-rdfizer. Docker Hub image, tag 3.3.0. Accessed 2026-10-07 (2026). <https://hub.docker.com/r/ecrum19/vcf-rdfizer>
- [57] Bolleman, J.T., Mungall, C.J., Strozzi, F., Baran, J., Dumontier, M., Bonnal, R.J.P., Buels, R., Hoehndorf, R., Fujisawa, T., Katayama, T., Cock, P.J.A.: FALDO: a semantic standard for describing the location of nucleotide and protein feature annotation. *Journal of Biomedical Semantics* **7**(1), 39 (2016) <https://doi.org/10.1186/s13326-016-0067-z>
- [58] Eilbeck, K., Lewis, S.E., Mungall, C.J., Yandell, M., Stein, L., Durbin, R., Ashburner, M.: The Sequence Ontology: a tool for the unification of genome annotations. *Genome Biology* **6**(5), 44 (2005) <https://doi.org/10.1186/gb-2005-6-5-r44>
- [59] Menotti, L., Cazzaro, M., Rueda, M., Gut, I.G., Silvello, G.: HERO-Genomics: Bridging Genomic Data and Ontological Modelling. In: *Semantic Web Applications and Tools for Health Care and Life Sciences (SWAT4HCLS 2025)*. CEUR Workshop Proceedings, Barcelona, Spain (2025). https://www.dei.unipd.it/~silvello/papers/2025_MCS_SWAT4HCLS.pdf

- [60] Wagner, A.H., Babb, L., Alterovitz, G., Baudis, M., Brush, M., Cameron, D.L., Cline, M., Griffith, M., Griffith, O.L., Hunt, S.E., Kreda, D., Lee, J.M., Li, S., Lopez, J., Moyer, E., Nelson, T., Patel, R.Y., Riehle, K., Robinson, P.N., Rynearson, S., Schuilenburg, H., Tsukanov, K., Walsh, B., Konopko, M., Rehm, H.L., Yates, A.D., Freimuth, R.R., Hart, R.K.: The GA4GH Variation Representation Specification: A computational framework for variation representation and federated identification. *Cell Genomics* **1**(2), 100027 (2021) <https://doi.org/10.1016/j.xgen.2021.100027>
- [61] Wagner, A.H., Hart, R., Babb, L., Kuzma, K., Schuilenburg, H., Stevenson, J., O'Neill, T., Epic, M.-T., Holmes, B., Moyer, E., Baudis, M., Robinson, P., Cline, M., Patel, R.Y., Freimuth, R.R.: ga4gh/vrs 2.0.1. Zenodo (2025). <https://doi.org/10.5281/zenodo.15059674> . <https://doi.org/10.5281/zenodo.15059674> Accessed 2026-03-02
- [62] De Meester, B., Heyvaert, P., Delva, T.: RDF Mapping Language (RML) (2024). <https://rml.io/specs/rml/> Accessed 2025-11-24
- [63] Oo, S.M., Haesendonck, G., De Meester, B., Dimou, A.: RMLStreamer-SISO: An RDF Stream Generator from Streaming Heterogeneous Data. In: Sattler, U., Hogan, A., Keet, M., Presutti, V., Almeida, J.P.A., Takeda, H., Monnin, P., Pirrò, G., d'Amato, C. (eds.) *The Semantic Web – ISWC 2022* vol. 13489, pp. 697–713. Springer, Cham (2022). https://doi.org/10.1007/978-3-031-19433-7_40 . Series Title: *Lecture Notes in Computer Science*. https://link.springer.com/10.1007/978-3-031-19433-7_40 Accessed 2025-11-24
- [64] Fernández, J.D., Martínez-Prieto, M.A., Gutiérrez, C., Polleres, A., Arias, M.: Binary rdf representation for publication and exchange (hdt). *Web Semantics: Science, Services and Agents on the World Wide Web* **19**, 22–41 (2013)
- [65] Arenas-Guerrero, J., Ferrada, S.: COTTAS: Columnar triple table storage for efficient and compressed RDF management. In: *The Semantic Web – ISWC 2025. Lecture Notes in Computer Science*, vol. 16141, pp. 313–331. Springer, Cham (2026). https://doi.org/10.1007/978-3-032-09530-5_18 . https://doi.org/10.1007/978-3-032-09530-5_18
- [66] Alakuijala, J., Farruggia, A., Ferragina, P., Kliuchnikov, E., Obryk, R., Szabadka, Z., Vandevenne, L.: Brotli: A General-Purpose Data Compressor. *ACM Transactions on Information Systems* **37**(1), 1–30 (2019) <https://doi.org/10.1145/3231935>
- [67] Bast, H., Buchhold, B.: QLever: A query engine for efficient SPARQL+Text search. In: *Proceedings of the 2017 ACM on Conference on Information and Knowledge Management (CIKM)*, pp. 647–656 (2017). <https://doi.org/10.1145/3132847.3132921> . <https://doi.org/10.1145/3132847.3132921>
- [68] Taelman, R., Herwegen, J.V., Sande, M.V., Verborgh, R.: Comunica: A modular

- SPARQL query engine for the web. In: Proceedings of the 17th International Semantic Web Conference, pp. 239–255. Springer, Cham (2018). https://doi.org/10.1007/978-3-030-00668-6_15 . <https://comunica.dev/>
- [69] Crum, E.: Comunica SPARQL COTTAS: a Comunica engine for query evaluation over COTTAS files. Version 0.1.0; npm package `@elias.crum/query-sparql-cottas` (2026). <https://github.com/ecrum19/comunica-feature-cottas>
 - [70] Danecek, P., Auton, A., Abecasis, G., Albers, C.A., Banks, E., DePristo, M.A., Handsaker, R.E., Lunter, G., Marth, G.T., Sherry, S.T., McVean, G., Durbin, R., 1000 Genomes Project Analysis Group: The variant call format and VCFtools. *Bioinformatics* **27**(15), 2156–2158 (2011) <https://doi.org/10.1093/bioinformatics/btr330>
 - [71] Pedersen, B.S., Quinlan, A.R.: cyvcf2: fast, flexible variant analysis with Python. *Bioinformatics* **33**(12), 1867–1869 (2017) <https://doi.org/10.1093/bioinformatics/btx057>
 - [72] World Wide Web Consortium: Shapes Constraint Language (SHACL). W3C Recommendation (2017). <https://www.w3.org/TR/shacl/> Accessed 2026-10-02
 - [73] Personal Genome Project (2005). https://my.pgp-hms.org/public_genetic_data Accessed 2025-11-24
 - [74] NIST Genome in a Bottle (2025). <https://www.nist.gov/programs-projects/genome-bottle> Accessed 2025-11-24
 - [75] The 1000 Genomes Project Consortium: A global reference for human genetic variation. *Nature* **526**(7571), 68–74 (2015) <https://doi.org/10.1038/nature15393>
 - [76] Ebert, P., Audano, P.A., Zhu, Q., Rodriguez-Martin, B., Porubsky, D., Bonder, M.J., Sulovari, A., Ebler, J., Zhou, W., Serra Mari, R., *et al.*: Haplotype-resolved diverse human genomes and integrated analysis of structural variation. *Science* **372**(6537), 7117 (2021) <https://doi.org/10.1126/science.abf7117>
 - [77] Wagner, J., Olson, N.D., Harris, L., Khan, Z., Farek, J., Mahmoud, M., Stankovic, A., Kovacevic, V., Yoo, B., Miller, N., *et al.*: Benchmarking challenging small variants with linked and long reads. *Cell Genomics* **2**(5), 100128 (2022) <https://doi.org/10.1016/j.xgen.2022.100128>
 - [78] Miller, D.T., Lee, K., Abul-Husn, N.S., Amendola, L.M., *et al.*: ACMG SF v3.2 list for reporting of secondary findings in clinical exome and genome sequencing: A policy statement of the American College of Medical Genetics and Genomics (ACMG). *Genetics in Medicine* **25**(8), 100866 (2023) <https://doi.org/10.1016/j.gim.2023.100866>

- [79] Byrska-Bishop, M., Evani, U.S., Zhao, X., *et al.*: High-coverage whole-genome sequencing of the expanded 1000 Genomes Project cohort including 602 trios. *Cell* **185**(18), 3426–344019 (2022) <https://doi.org/10.1016/j.cell.2022.08.004>
- [80] National Center for Biotechnology Information: ClinVar VCF Release (GRCh38). Header declares `##fileformat=VCFv4.1`; accessed 8 September 2026 (2026). https://ftp.ncbi.nlm.nih.gov/pub/clinvar/vcf_GRCh38/
- [81] Anthropic: Claude Opus 5.5 (Extra) [Large language model]. Accessed through the Claude Code desktop application. Used September 2026 (2026). <https://www.anthropic.com/claude>
- [82] OpenAI: GPT-6 Astra (X-High) [Large language model]. Accessed through the Codex desktop application. Used September 2026 (2026). <https://openai.com/>